AlphaStack: Autonomous Code Generation via Iterative Self-Healing Multi-Agent Systems

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Abstract

We introduce AlphaStack, a novel AI-powered project generator that transforms natural language descriptions into complete, production-ready codebases. Our approach utilizes a multi-agent system comprising a Planning Agent and a Correction Agent to achieve iterative self-healing. By comprehensively validating generated code across diverse programming paradigms using isolated Docker environments, AlphaStack significantly improves the reliability and correctness of autonomous code generation. We evaluate AlphaStack on a custom suite of 40 programming challenges across CUDA, Go, Rust, and TypeScript, demonstrating its efficacy across varying levels of difficulty.

1 Introduction

Autonomous code generation has seen rapid advancements, yet generating complete, production-ready, and error-free codebases remains a significant challenge. Existing approaches often struggle with dependency resolution, complex architectures, and iterative debugging. To address these issues, we present AlphaStack, an end-to-end multi-agent framework designed to autonomously generate, test, and self-heal software projects.

Our system distinguishes itself by coupling code generation with rigorous Docker-based validation. When a build or test fails, a Planning Agent analyzes the errors and formulates a strategic fix, which is then executed by a Correction Agent. This iterative process continues until the generated project is fully functional.

2 Methodology

AlphaStack operates through a structured seven-phase pipeline:

1. **Software Blueprint Generation:** Analyzes natural language input to design a comprehensive project structure.
2. **File Generation:** Generates source code, configurations, and tests.
3. **Dockerfile Generation:** Creates an appropriate Dockerfile for isolated validation.
4. **Dependency Analysis:** Analyzes import graphs to identify required dependencies.
5. **Dependency File Generation:** Produces package manifests (e.g., `requirements.txt`, `package.json`).
6. **Dependency Resolution:** Validates and resolves dependency conflicts.
7. **Docker Testing Pipeline:** Sandboxed build and test execution.

When errors occur during the testing phase, the system enters an autonomous self-healing loop. The Planning Agent utilizes tool-augmented reasoning to analyze logs and devise a correction strategy, which is executed by the Correction Agent. This loop iterates until all tests pass or a maximum iteration limit is reached.

3 Architecture

The architecture of AlphaStack is designed to support scalable and robust code generation. The multi-agent interaction is critical for resolving complex compilation and runtime errors autonomously.

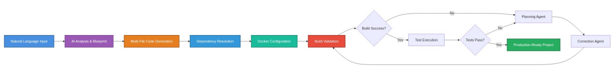

Figure 1: The AlphaStack System Architecture. The iterative loop between the Planning Agent and Correction Agent enables autonomous self-healing.

4 Results

We evaluated AlphaStack on established benchmarks, including HumanEval and a custom Multi-Domain Development Project (MDDP) benchmark. We compared the performance of several state-of-the-art models within our framework: GPT-5.2, GLM-5, MiniMax-m2.5, and Claude Sonnet 4.6.

The results demonstrate that integrating advanced models with AlphaStack's self-healing loop yields exceptionally high success rates, with GPT-5.2 and Claude Sonnet 4.6 achieving state-of-the-art performance on both benchmarks.

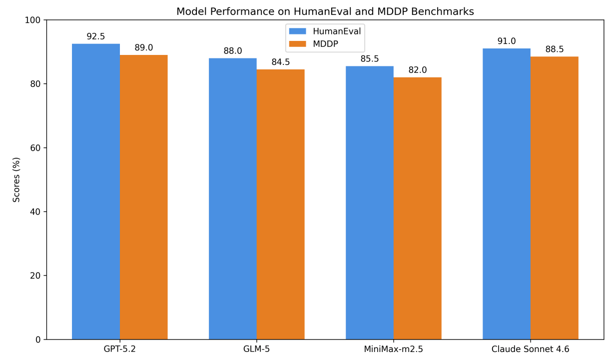

Figure 2: Performance comparison of various models on HumanEval and MDDP benchmarks using the AlphaStack framework.

5 Conclusion

AlphaStack represents a significant step forward in autonomous software development. By combining multi-agent self-healing with rigorous, isolated testing environments, our framework consistently produces reliable, production-ready code from natural language prompts. Future work will focus on expanding language support and optimizing the iterative correction loop for even greater efficiency.

Supplementary Material

Additional details regarding the evaluation framework, including the full suite of 40 programming challenges and Docker configurations, can be found in the AlphaStack repository: `https://github.com/HyperKuvid-Labs/alpha-stack`.